

Anastasios Vlachos

✉ a.vlachos@rug.nl · 🌐 [Personal Website](#) · 🎓 [Google Scholar](#)

EDUCATION

University of Groningen PhD in Systems, Control and Optimization – Supervisor: Prof. dr. ir. Bart Besselink – Topic: Contract-Based Verification for Modular Control System Design	Groningen, Netherlands Sep. 2025 - present
ETH Zürich Master of Science in Information Technology and Electrical Engineering – Specialization: Systems and Control – GPA: 5.4/6 – Thesis: “The Price of Constraints in Linear Tracking Model Predictive Control”, Grade: 6/6	Zürich, Switzerland Sep. 2022 - Apr. 2025
National Technical University of Athens (NTUA) Diploma in Electrical and Computer Engineering (5-year degree; 300 ECTS) – Major: <i>Electronics & Systems</i> – GPA: 8.94/10.00 (top 8%) – Thesis: “Mathematical Analysis of $\Sigma\Delta$ Modulators and Applications”, Grade: 10/10	Athens, Greece Sep. 2015 - May 2021

RESEARCH EXPERIENCE

Automatic Control Laboratory (IfA), ETH Zürich Master Thesis – Worked on studying the influence of constraints on Linear Quadratic Tracking problems. – Provided theoretical complexity bounds that captured the aforementioned influence. – Conducted numerical simulations that verify these bounds. – Supervised by Aren Karapetyan, Dr. Anastasios Tsiamis, Dr. Efe Balta and Prof. Dr. John Lygeros.	Zürich, Switzerland Sep. 2024 - Apr. 2025
Automatic Control Laboratory (IfA), ETH Zürich Semester Project – Worked on bridging offline and online prediction methods in the problem of Pedestrian Tracking. – Proposed a method to adapt online to the residual errors of offline-trained models. – Provided Regret guarantees for our method and conducted simulations to showcase its efficacy. – Supervised by Dr. Anastasios Tsiamis, Aren Karapetyan, Dr. Efe Balta and Prof. Dr. John Lygeros.	Zürich, Switzerland Oct. 2023 - Mar. 2024
Circuits and Systems Group (Prof. P.P. Sotiriadis), NTUA Diploma Thesis - Undergraduate Research Assistant – Completed my diploma thesis by examining $\Sigma\Delta$ modulators from a dynamical-systems viewpoint and presenting their usefulness in Frequency Synthesis and Stochastic Filtering applications. – Resulted in publication at MOCAST 2021 (nominated for Best Paper Award) and an invited journal paper.	Athens, Greece Mar. 2020 - Dec. 2021

PUBLICATIONS

- [1] **A. Vlachos**, A. Tsiamis, A. Karapetyan, E. C. Balta, and J. Lygeros, “Online residual learning from offline experts for pedestrian tracking,” *arXiv preprint arXiv:2409.04069*, 2024.
- [2] N. Temenos, **A. Vlachos**, and P. P. Sotiriadis, “Efficient stochastic computing fir filtering using sigma-delta modulated signals,” *Technologies*, vol. 10, no. 1, p. 14, 2022.
- [3] **A. Vlachos**, N. Temenos, and P. P. Sotiriadis, “Exploring the effectiveness of sigma-delta modulators in stochastic computing-based fir filtering,” in *2021 10th International Conference on Modern Circuits and Systems Technologies (MOCAST)*, IEEE, 2021, pp. 1–4.

TEACHING EXPERIENCE

Linear Systems Theory (ETH Zürich) Teaching Assistant – Consult on the clarity, correctness and difficulty of the proposed homework and grade the homework submissions of students, on a biweekly basis.	Autumn 2023, 2024
Advanced Topics in Control - Distributed Systems and Control (ETH Zürich) Teaching Assistant – Grade the homework submissions of students, on a monthly basis, and provide grades statistics.	Spring 2024

TECHNICAL SKILLS

Programming Languages: Python, Matlab, C/C++
Software Frameworks: PyTorch, Scikit-Learn, NumPy,
Simulation Tools: ROS-Gazebo
Other Tools: Latex

LANGUAGES

English(fluent, C2), **German**(intermediate, B1), **Greek**(native)